

K-Comp V2 – a brief overview

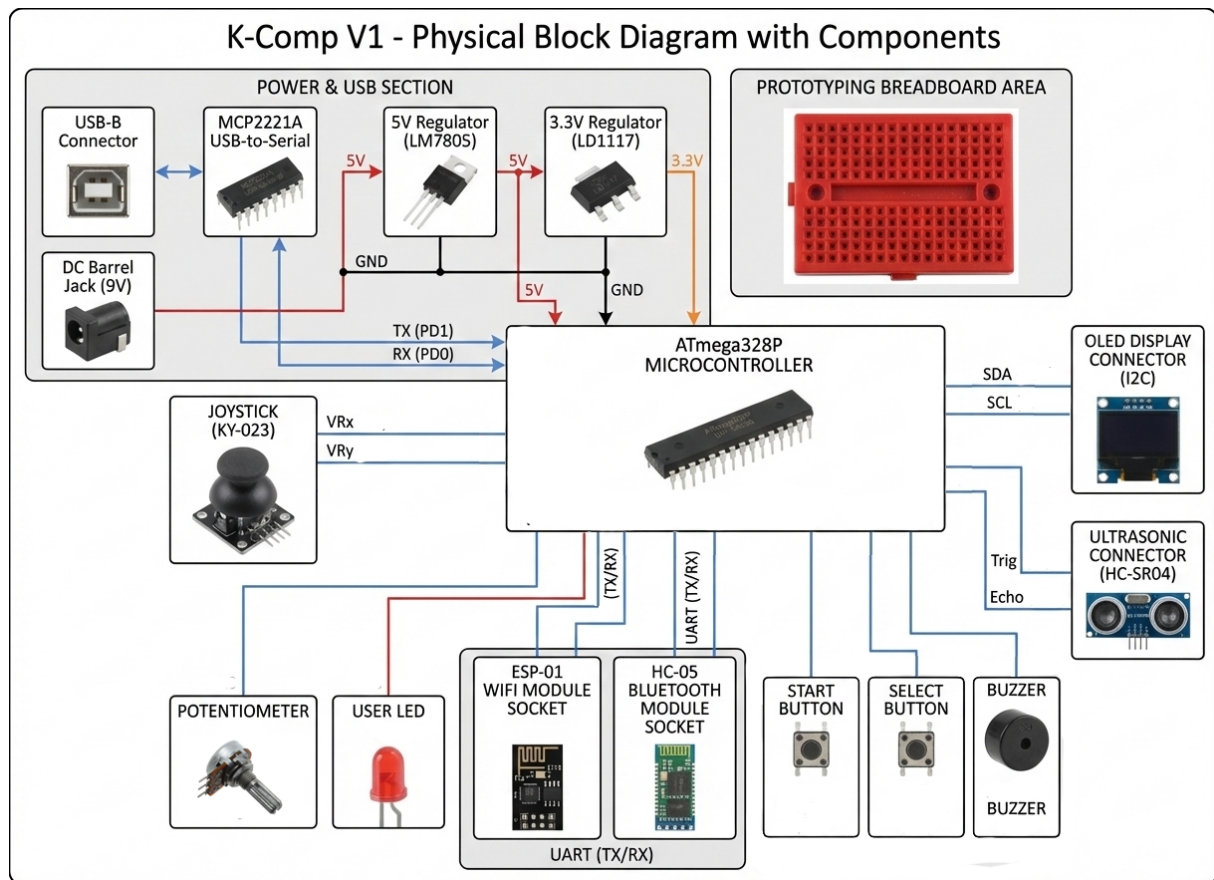
The *K-Comp V2* is a custom-designed *single board computer (SBC)* tailored for learners of the C programming language working on interactive hardware projects. It is based on the *ATmega328P* microcontroller, the same chip used on Arduino Uno boards, but it extends the functionality with pre-connected components, custom made libraries and is fully supported in the Arduino IDE. It should be suitable for learning to develop embedded systems and sensor applications in C with minimal wiring and setup.

Hardware Specifications of the ATmega328P

Feature	Value	Explanation
Controller	ATmega328P	The main processor that runs your code and controls all components.
Clock Speed	16 MHz	Determines how fast the microcontroller can execute instructions (e.g. ADD, LOAD, JMP, ...) 16 million cycles per second).
Data Width	8 bit	The processor works with 8 bits at a time – for example, it adds or stores data in 8-bit chunks.
Flash Memory	32 KB	Stores your program permanently, even when power is off.
SRAM	2 KB	Temporary memory used to store variables while the program is running.
EEPROM	1 KB	A type of memory where you can save data permanently from within your code (e.g., settings).
USB Interface	MCP2221A	Converts USB signals from the PC to serial data the microcontroller can understand.
Power Supply	USB (or opt. 5V external)	The board can be powered directly via USB (or optionally from a separate 5V power source).
Digital I/O Pins	14	Used to read digital sensors or control devices like LEDs, buzzers, or relays.
Analog Inputs	6	For reading continuous signals (e.g., from a potentiometer or joystick).

Main Components (All Prewired):

(only gives the basic idea!)



🧠 Main Processor – ATmega328P

- This is the heart of the system — it runs your Code
- Digital pins (D0–D13) for controlling LEDs, buzzers, buttons, etc.
- Analog inputs (A0–A5) for reading from sensors like the potentiometer or joystick.
- It gets programmed via the MCP2221A.
- A 16 MHz crystal oscillator provides the timing.

● Red LED

- For feedback, indicators, or status signals.
- Connected to D3.

🔌 USB Communication – MCP2221A

- The MCP2221A (IC2) is a USB-to-serial chip. It connects the board to your PC via USB and takes care of:
- Uploading code to the ATmega328P
- Serial communication between the PC and the microcontroller
- It connects to the TX (D1) and RX (D0) lines of the ATmega328P.

○ 2 Push Buttons

- For user input, program control, or state switching.
- Connected to D7 (START) and D8 (SELECT).

 Potentiometer - Analog input for user-adjustable values (e.g., brightness, delay, volume). - Connected to A0 for analog input.	 Passive Buzzer - Can generate tones for alarms or sound output. - Connected to D5.
 Joystick (KY-023) - Two analog axes (X, Y). - For navigation, menus, or control inputs. - X and Y axes go to A1 and A2, so they're read as analog values.	 0.96" OLED Display (SSD1306, I²C) 128×64 pixels. - Display text, values, or simple graphics. - Controlled via I2C lines: SCL (PC5) and SDA (PC4) on the ATmega328P.
 HC-SR04 Ultrasonic Distance Sensor - Measures distances between 2 cm and 400 cm. - Trigger pin to D12, echo pin to D11.	 ESP-01 WiFi Header (Optional) Lets you add an ESP-01 WiFi module for wireless communication.
 Pin Headers and Jumpers You can use jumpers (JP_RESET, JP_SEL, JP_STA, etc.) to enable or disable parts of the circuit, or for debugging.	 HC-05 Header (Optional) Lets you add an HC-04 Bluetooth module for wireless communication.

Software and Development Environment

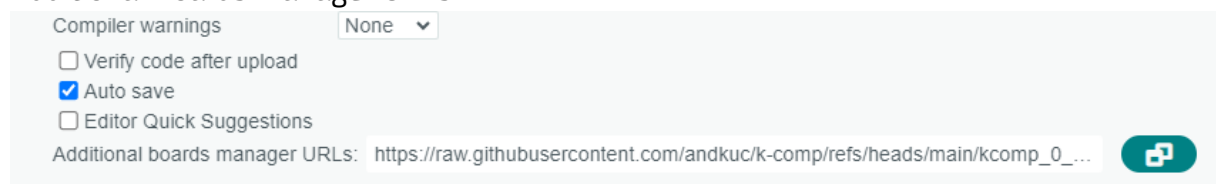
- Fully supported in the **Arduino IDE**, **Visual Studio Code (PlatformIO)** and **MBlock 5**.
- Comes with a **custom board package**, **libraries**, and **ready-to-run examples**
- Libraries abstract away complex hardware access — focus on logic, not details

a) Using K-Comp in the Arduino IDE

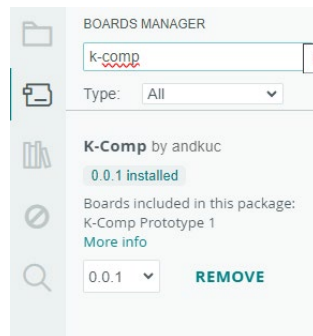
This section shows how to add K-Comp V2 to your Arduino IDE and to compile and upload sample programs.

1) Install Arduino IDE (<https://www.arduino.cc/en/software>) and start it.

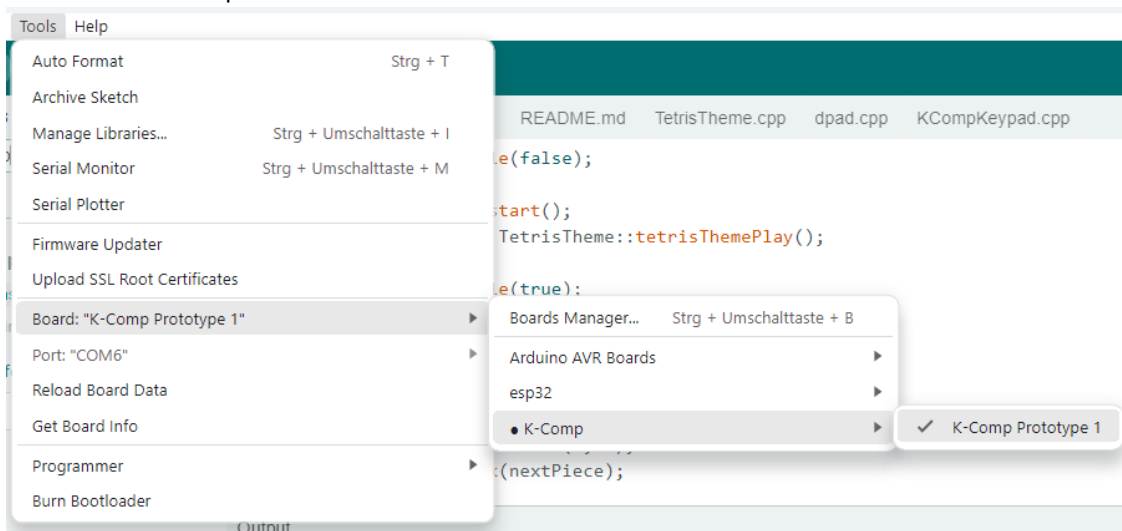
Open File | Preferences and add https://raw.githubusercontent.com/andkuc/k-comp/refs/heads/main/board-definition/kcompV2/kcompV2_1_0_0_package_index.json to Additional Boards Manager URLs.



- 2) Open the Boards Manager on the left side of the Arduino IDE and install K-Comp. Do not forget to install the driver for MCP2221A!

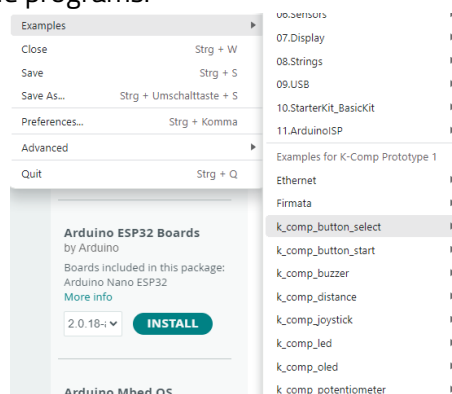


- 3) Select K-Comp V1 under Tools | Board.

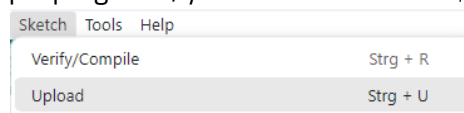


- 4) Select the COM port if K-Comp is connected!

- 5) Try out some of the sample programs:



In order to upload the sample programs, you select Sketch | Upload



Note: If you run Linux, make sure to add the latter udev rule for MCP2221A if you want to make use of auto-reset before upload!

```
#!/bin/bash
# K-Comp Board Rule Installer
# Run this once with sudo: sudo ./install_linux_rules.sh

echo "Installing udev rules for K-Comp (MCP2221A)..."
```

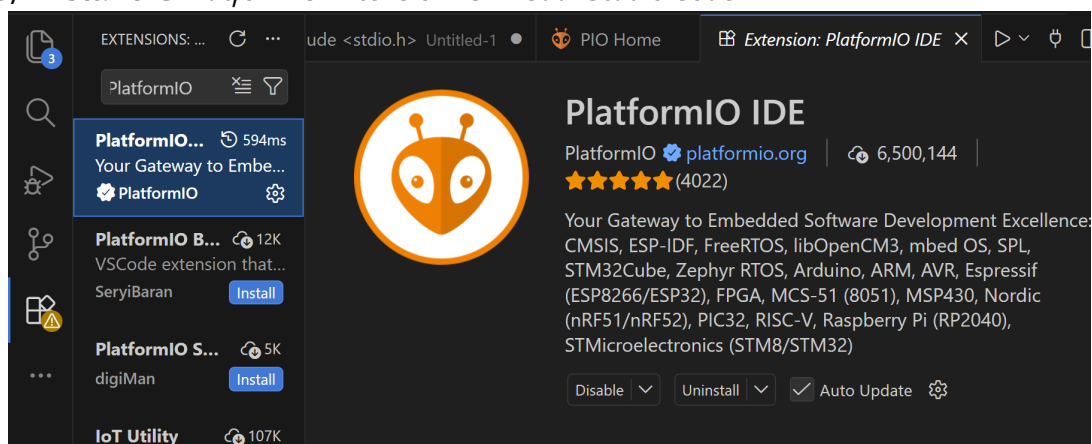
```
# 1. Create the rule file
echo 'SUBSYSTEM=="usb", ATTRS{idVendor}=="04d8", MODE="0666",
GROUP="plugdev"' | sudo tee /etc/udev/rules.d/99-kcomp.rules > /dev/null

# 2. Reload rules
sudo udevadm control --reload-rules
sudo udevadm trigger

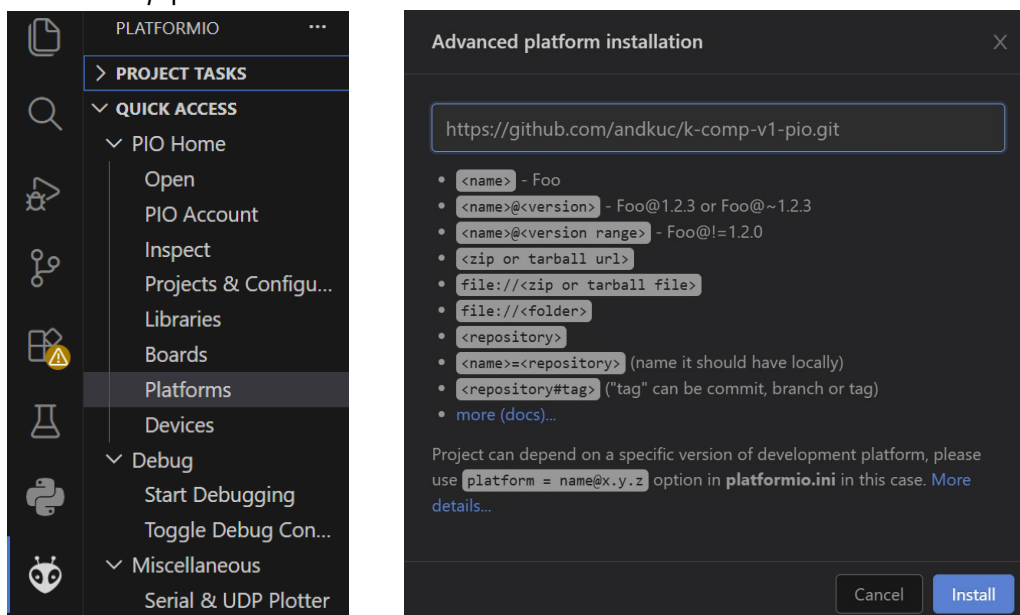
echo "Success! Please unplug and replug your K-Comp board."
```

b) Using K-Comp in Visual Studio Code

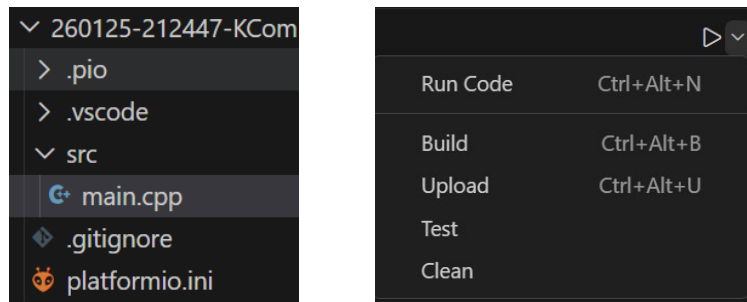
- 1) Make sure to have git installed (<https://git-scm.com/install>).
- 2) Install the latest version of *Visual Studio Code* (<https://code.visualstudio.com>).
- 3) Install the *PlatformIO Extension for Visual Studio Code*:



- 4) Open the *PlatformIO Extension* and navigate to *Platforms* and select *Advanced Installation*. Enter <https://github.com/andkuc/k-comp-v1-pio.git> and install the *K-Comp* platform.



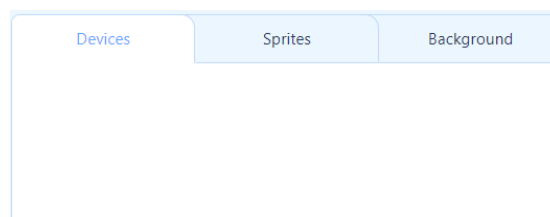
- 5) Navigate to *PIO Home* and select *Open Project Examples*. Pick a *K-Comp* example project of your choice and import it.
- 6) Navigate to `main.cpp`. You can build and upload the example project like in the screenshot:



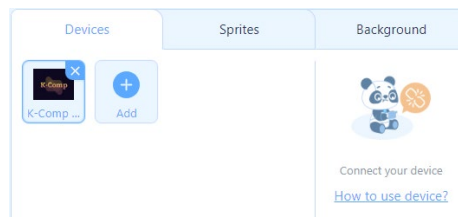
- 7) You can open the Serial Monitor by clicking on the  symbol.

c) Using K-Comp in Mblock 5

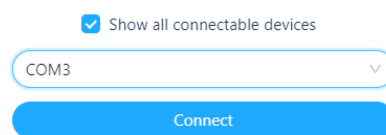
- 1) Download and install *Mblock 5* (<https://mblock.cc/pages/downloads>).
- 2) Run *Mblock 5* and delete all devices.



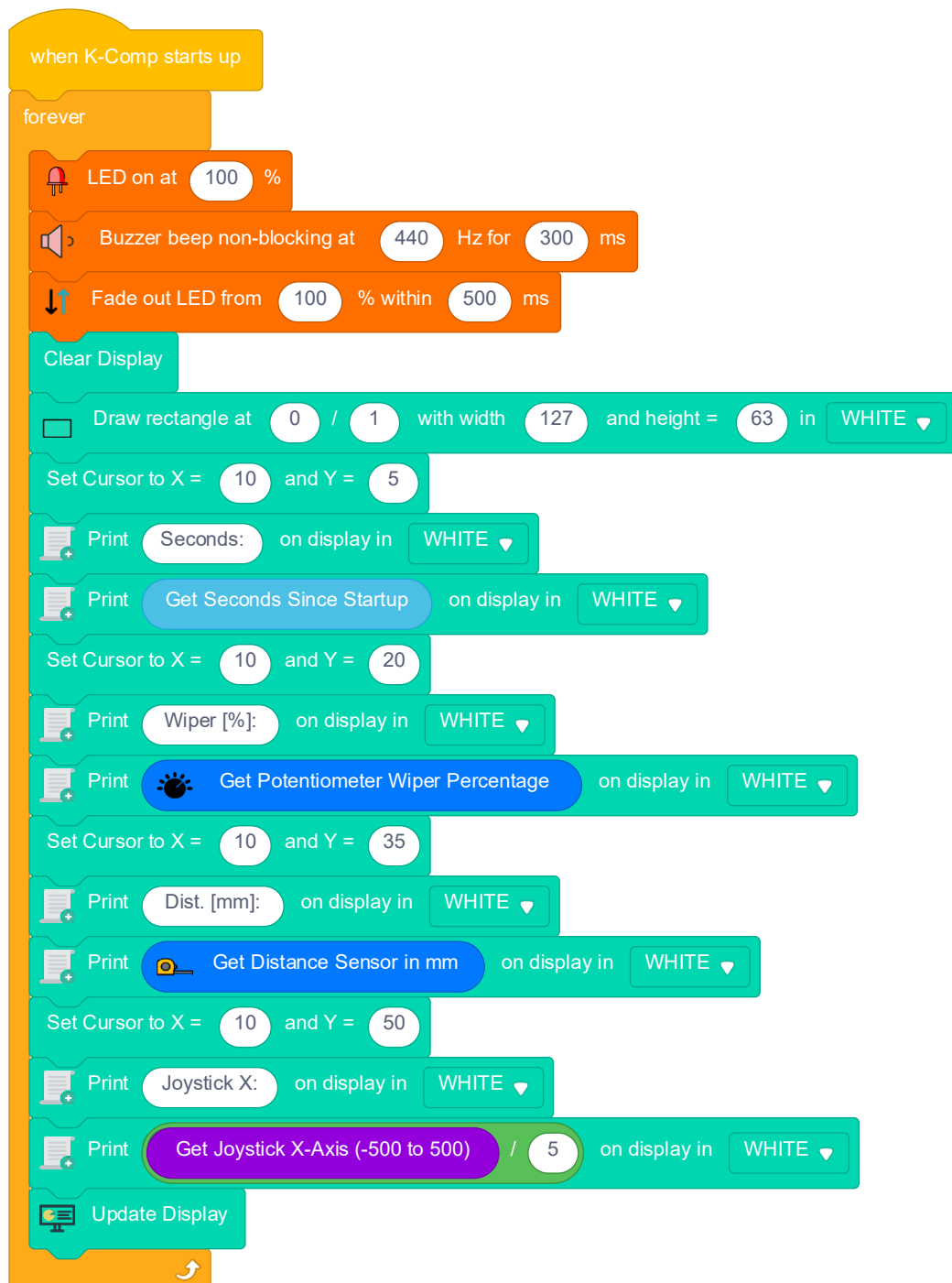
- 3) Download `k_comp_v1.mext` from the *K-Comp Mblock repository* (<https://github.com/andkuc/k-comp-mblock5>).
- 4) Drag `k_comp_v1.mext` into the *Mblock 5 IDE*.



- 5) Connect to *K-Comp*. Make sure you have `Show all connectable devices` selected.



- 6) You now can program *K-Comp* by arranging blocks! See the example below:



```

Seconds: 25
Wiper [%]: 37
Dist. [mm]: 127
Joystick X: -123
  
```

- 7) Press Upload code and upload the program. **Press and release the Reset-button on the K-Comp board as soon as start uploading... . shows up** in the Upload progress dialog!